Fifty Dimensions for Ten Thousand Senses: How a Transformer Packs More Senses than Width

Alexandru-Mihai Negrescu
negrescualex05@gmail.com

Abstract

A one-layer transformer is trained on a synthetic language in which 10^4 word senses, with known Zipf frequencies and known cue-based co-activation, must be read out of a d=4096 residual stream. The width leaves room for a near-orthogonal code, and the question is what SGD does with that room. It compresses. The sense code occupies 46–50 effective dimensions after training (42 after a split-half noise correction), about 1% of the width, and a spectrum-matched null shows that the code's bulk non-orthogonality is this dimensionality and nothing more. The one structured departure is a tail of sense pairs that share context cues. That tail is inherited from the input: it is present and graded in cue overlap at initialization, because a bag-of-tokens layer sums the cue embeddings into the residual, and training amplifies it along with the compression while keeping the shared-to-disjoint ratio near 10×. A registered rank-mixed corpus shows cue identity carries most of it and frequency rank about 5°; the second half of that rule, a subspace separation, failed. A width sweep from 64 to 4096 shows the task saturates at 0.9 accuracy with 128 dimensions and a ten-dimensional code, while the code SGD builds keeps growing with the width, about $d^{0.45}$, for no gain in accuracy; the tail is flat throughout. An interference penalty spreads the bulk to the null level without removing the tail. Two measurement pitfalls are results in their own right: per-word centering forces the same-word cosine to $-1/(k-1)$, and an isotropic null in $\mathbb{R}^d$ misreads low dimensionality as failed packing. The design was pre-registered; every failed rule is reported.

1 Introduction

A layer of width d can hold far more than d features if their directions are only approximately orthogonal, and the number of unit vectors with pairwise $|\cos| \leq \sin\varepsilon$ grows as $e^{c\varepsilon^2 d}$ (Tao, 2013; Garg et al., 2026). Superposition theory (Elhage et al., 2022; Scherlis et al., 2022) takes this as the mechanism by which networks represent more features than neurons, and at saturation it predicts tight frames for independent sparse features, with correlated features allowed to collapse or to share directions. The predictions have been tested on autoencoders with independent sparse features. They have not been tested on a next-token model trained on a language whose feature inventory, frequencies, and co-activation structure are known and exceed the width. Our features are correlated by construction, so the tight-frame prediction is not the one at stake. The question is what SGD does with the near-orthogonal headroom when it has it. The answer is that it compresses, in line with the low-rank bias of gradient training (Papyan et al., 2020; Ansuini et al., 2019; Huh et al., 2023), and the only structured departure from that compressed code is one the input already contains.

Polysemy is the natural test case. Senses of one word never fire together, so theory lets them share a subspace at no cost; senses of different words co-activate, so that is where interference binds; frequency sets the allocation. We build a language in which all three are controlled, train a transformer with S=10^4 senses in d=4096, and ask what SGD packs.

Contributions.

1. SGD compresses the sense code. The protocol centroids have participation ratio 46–50 of 4096 after training (Gaussian reference 2905; untrained initialization 383, itself an upper bound since centroid noise inflates PR), and a spectrum-matched null shows the code's bulk non-orthogonality is this dimensionality, p50 5.36°–5.51° null against 5.41°–5.65° observed. The tail is what remains, p99.9 54.2°–54.8° against 25.8°–26.8° (§4.2).
2. The tail's pattern is inherited from the input; training amplifies it. Shared-cue pairs are already aligned at initialization, median 17° against 1.6° for cue-disjoint pairs and graded in cue overlap 8° to 21°, as expected when a bag-of-tokens layer sums the cue embeddings into the residual. After training the two populations sit at 54° against 5.6°, the same 10× ratio, with the shared-cue $|\cos|$ raised from 0.29 to 0.80. A registered rank-mixed corpus puts most of the alignment on cue identity and about 5° on frequency rank, also present at initialization; the rule's subspace clause failed (§4.3).
3. A width sweep from 64 to 4096 shows width buys dimensions, not accuracy. At the better of two learning rates, accuracy is 0.87 at d=64 and 0.90–0.93 from d=128 up, while the code's participation ratio grows from 8 to 50 (7 to 42 after split-half noise correction), about $d^{0.45}$, and the shared-cue tail is width-invariant. The registered sweep at a single learning rate had read the rate limit as a capacity limit (§4.2).
4. Two interventions bound the tail. An interference penalty, priced by co-activation or uniformly, spreads the bulk to the null level (ratio 0.96–0.97) and lowers the tail by about 2° without removing it (§4.4); readout deletion of the code's top directions is directional but a registered fairness check failed, every corpus-informed basis coincides with the shared one, and no mechanism is named (§4.5).

The design, endpoints, and decision rules of every run were written down before execution (Appendix A). Pre-registered are the capacity endpoints, the fairness check, the width sweep, and the rank-mixed corpus. Post-hoc are the decomposition, the nulls, the initialization baselines, and the centering identity, each labeled as such. Two estimator pitfalls we hit are stated as results (§4.1, §4.2): per-word centering forces the same-word cosine to $-1/(k-1)$, and an isotropic null in $\mathbb{R}^d$ misreads low dimensionality as failed packing.

2 Setup

Language. W ambiguous words, each with k senses at Zipf frequencies $P(r) \propto 1/r$. Sentences are 2–4 chunks of cue word [cont] filler; exactly one chunk is the main slot and carries a continuation token $\mathrm{cont}_{w,r}$ determined by (cue, word) and by nothing less: the cue alone leaves 8–20 candidates, the word alone leaves k. One word appears at most once per sentence, so same-word senses never co-activate. Cues are drawn from pools indexed by (block, rank); senses in the same block and rank share a pool of 4 cues, so cross-word co-activation structure is set by the block assignment. At the capacity scale, W=2000, k=5 (S=10^4), 500 blocks, 96,000 sentences, vocabulary 14,082; blocks are rank-pure by construction, so every shared-cue pair is same-rank. A 10% held-out split is used for behavior. Structural checks (Zipf recovery, zero same-word co-occurrence, deterministic (cue, word)→cont, no duplicates or split leakage, prior-only ceiling 0.438) are in Appendix D.

Model. One block: single-head causal attention, ReLU MLP ($4d$), untied embedding/unembedding, no biases, no LayerNorm (so AdamW (Loshchilov & Hutter, 2019) weight decay γ keeps the semantics of Liu et al. (2025)'s superposition knob), no positional code (chunk order carries no signal). Loss is masked to continuation positions. d=64 for the toy regime (W=12, k=4, 2500 steps, 5 seeds per configuration); d=4096 (317M parameters) for the capacity regime, 6000 steps, batch 256, lr 10^{-4}, on one 8 GB consumer GPU (Appendix B lists the numerics amendments and their bitwise or bounded-delta gates).

Intervention. $\lambda > 0$ adds $\lambda\,\overline{\cos^2}$ between residual vectors at prediction positions whose targets belong to different words; same-word pairs are excluded so the penalty never fights the simplex. Under coact pricing, pairs enter at their empirical batch co-activation rate; under uniform, inverse-frequency weights $1/(f_i f_j)$ make every cross-word pair type's expected price equal, so rare shared-cue pairs are priced directly. Penalty statistics are always fp32.

Measurement. Sense vectors are residual-stream activations at the prediction position. Per-word centering precedes every angle (raw cosines are dominated by anisotropy and token identity (Ethayarajh, 2019); Ait Hou & Hwa (2026) show the same lexical confound in production models). Same-word geometry is classified by the pre-registered rule on the median cross-sense cosine (< -0.15 simplex, $|x| \leq 0.15$ orthogonal, > 0.15 collapsed; now shown ill-posed, §4.1). Split-half reliability of centroids is reported per rank as the denominator for every geometric claim. Cross-word interference is $\sum_{i<j} \cos^2(v_i, v_j)P(i, j)$ with P the full-corpus co-occurrence. Packing is measured on per-word-centered normalized centroids as angular deviation from 90° against a matched random baseline (same n, d; unit Gaussian directions), on a log-spaced ±4°–10° violation grid; no bound constant was precommitted; the SGD-vs-random gap is the quantity. Geometry uses the full corpus (at 48 sentences/word the test split leaves < 1 vector for rare senses); behavior is held-out.

3 Toy regime: the knobs work, and the rare senses merge

At d=64 (Appendix Table 6, Figure 1) weight decay is inert to γ=1, opens a graded window at $\gamma \in [1.5, 3]$ where accuracy holds and interference rises $0.32 \to 0.42$ as decay removes weakly supported directions first, and kills the model at γ=4. The two rarest senses of each word merge: r_3–r_4 is the least-negative of six pairs in 23/25 healthy runs (null 1/6), with a dose response $-0.05 \to +0.13$ while the other pairs sharpen to -0.41. The cross-word penalty abolishes the merge (0/5 runs at $\lambda \in \{1, 3\}$) while cutting interference $0.32 \to 0.25 \to 0.19$ at constant accuracy; γ paid 3 points at γ=3. A structural fact limits how far toy behavior can be read: with no positional code and one layer the output depends only on the multiset of prefix tokens, and in 17% of toy sentences a distractor cue forms a valid pair with the target word, so toy accuracy (0.91) sits at the order-blind ceiling (0.917) and toy per-rank behavior is meaningful between arms, not absolutely. At capacity scale the rate is 0.62% and the endpoints are clean (Appendix D).

4 Capacity regime: S=10^4 senses in d=4096

4.1 Same-word geometry: differentiation, and an estimator identity

The pre-registered rule classified the median per-word-centered same-word cosine, < -0.15 simplex, $|x| \leq 0.15$ orthogonal, > 0.15 collapsed. The rule was ill-posed. Centering k equal-norm equiangular vectors on their mean gives $-1/(k-1)$ whatever the raw geometry, so the estimator prints -0.25 for k=5 whenever the senses are exchangeable in norm, and an untrained model gives -0.249. The measured -0.250 [-0.250, -0.249] over the four seeds is the identity, and Table 1 keeps the column as an identity check.

What training does is differentiate the senses from the word component. The raw same-word cosine falls from 0.953 at initialization to 0.527 after training, and the per-word base-norm to sense-deviation-norm ratio falls from 4.90 to 1.26. The rank-pair matrix is graded at initialization (-0.10 at r_1r_2 to -0.34 at r_4r_5, a small-sample artifact of the rare-sense centroids) and uniform after training (-0.23 to -0.29, Figure 1), and the toy-scale r_3–r_4 merge is absent. Held-out accuracy 0.905 [0.900, 0.910], per-rank 0.98/0.95/0.86/0.76/0.64 for $r_1..r_5$ with reliability 0.89/0.79/0.67/0.56/0.49; r_5 centroids rest on ~ 4 sentences per word. Weighted interference is 0.0235.

4.2 Cross-word packing: fifty effective dimensions, and a tail

The frame-operator spectrum (Ivanov et al., 2026) of the per-word-centered sense centroids has participation ratio (PR) 46–50 of 4096 after training, down from 383 at initialization. A Gaussian code of the same shape gives 2905. The top 16 eigen-directions carry 45–47% of the variance (Appendix Table 7). Both PR values are upper bounds: centroid sampling noise pushes PR toward the Gaussian value, and it does so more at initialization. The compression is therefore at least severalfold, and its exact factor is not measured. The registered comparison was to an isotropic code in $\mathbb{R}^d$ (null A) on a ±4° violation grid, and SGD fails it outright, 0.62 of cross-word pairs outside ±4° against 0.00 for the matched

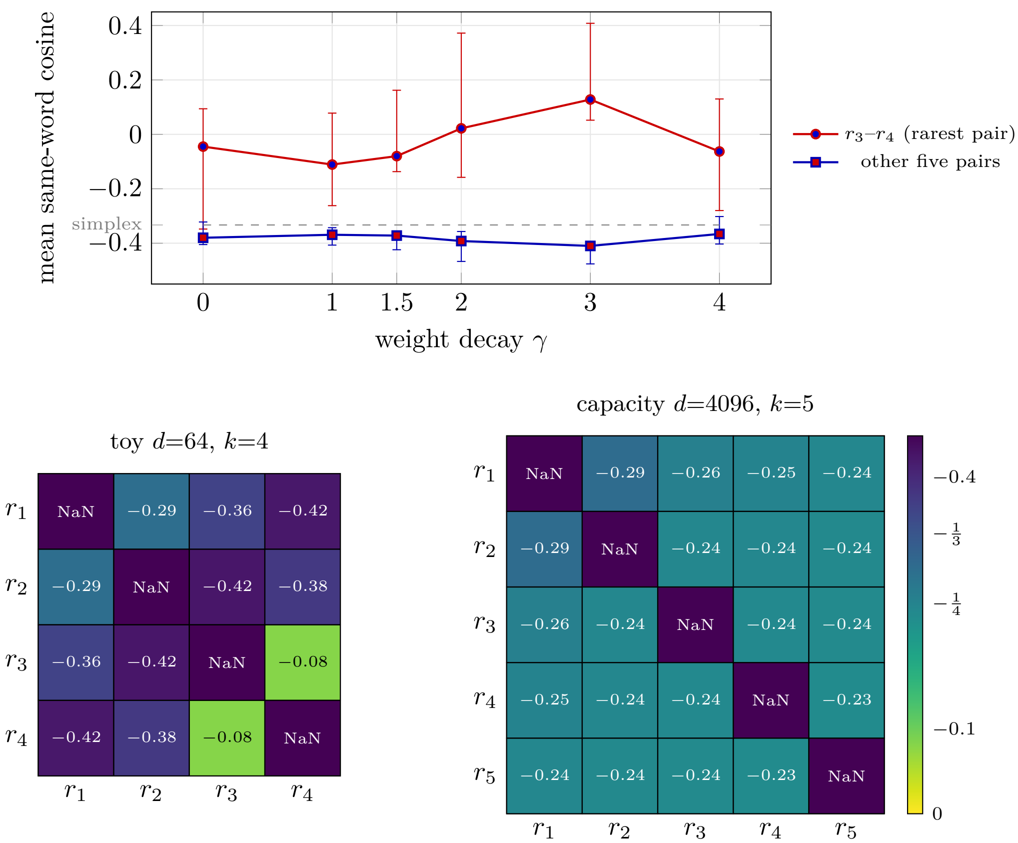

Figure 1: Same-word geometry. Top: toy dose response in γ; points are medians over 5 seeds, bars the seed range. The rarest pair drifts toward zero while the other five sharpen. Bottom: mean rank-pair cosine at γ=0. The toy r_3–r_4 merge is the one light cell (−0.08); at capacity every pair sits at the simplex value −0.25 (4 seeds, ranges $\leq$ 0.02). Colorbar marks the k=4 and k=5 simplex values.

Table 1: Capacity regime. Each cell: median over runs, with the seed range beneath. γ=0: 4 independent seeds plus a same-seed bf16 replicate (5 runs); λ=1 coact: seed 0 in fp32 and bf16 (2 runs); λ=1 uniform: 1 run. The same-word cosine is the per-word-centered estimator, whose −0.25 value is the centering identity and an identity check, not evidence. Deviation is the angular distance from 90° of cross-word centroid pairs (5.0×10^7 per run); viol is the registered endpoint, the fraction of pairs outside ±4° (matched random code: 0.0000 in every run). Null A (isotropic code in $\mathbb{R}^d$, the paper's original baseline) gives 0.60° / 2.94° / 5.2° (p50 / p99.9 / max), and null B (spectrum-matched) gives p50 5.36°–5.51°, p99.9 25.8°–26.8° for γ=0.

arm	acc	same-word cos	interf.	viol ±4°	p50	p99.9	max
γ=0	0.905 [0.900, 0.910]	−0.250 [−0.250, −0.249]	0.0235 [0.0234, 0.0257]	0.620 [0.619, 0.633]	5.44° [5.41, 5.65]	54.6° [54.2, 54.8]	73° [73, 74]
λ=1 coact	0.910 [0.904, 0.915]	−0.250 [−0.250, −0.249]	0.0153 [0.0147, 0.0158]	0.532 [0.525, 0.539]	4.32° [4.24, 4.39]	52.9° [52.8, 53.0]	72° [72, 72]
λ=1 uniform	0.908	−0.248	0.0150	0.521	4.21°	52.5°	71°

random code (Table 1). That comparison was the wrong one. Isotropic vectors in 46–50 dimensions predict a median deviation of 5.47°–5.68°, and the observed p50 is 5.41°–5.65°. A spectrum-matched Gaussian null (null B, $N(0, \Sigma)$ with the empirical centroid covariance, normalized; post-hoc) gives p50 5.36°–5.51°. This match is expected for any code with

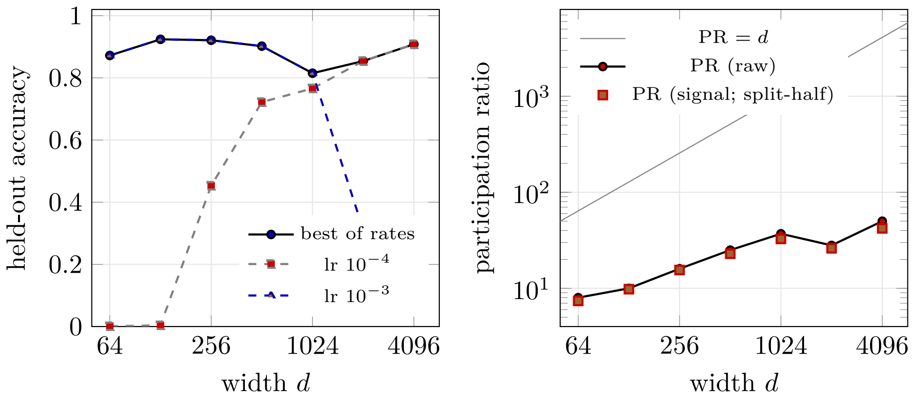

Figure 2: Width sweep, seed 0, bf16, same corpus and steps. Left, held-out accuracy at each of the learning rates tried and the better of them; the registered sweep (lr 10^{-4}, $d \geq 512$) was rate-limited. Right, participation ratio of the within-word code, raw and noise-corrected by split-half cross-covariance, against the width. The code grows about as $d^{0.45}$ while accuracy is flat.

that covariance, since the median pairwise cosine is pinned by the second moments; it says that the bulk carries no structure beyond its dimensionality, not that dimensionality was discovered (Appendix Table 8).

The tail carries structure. The observed p99.9 is 54.2°–54.8° against 25.8°–26.8° for null B, and the maximum is 73°–74° against 42°–45° (Appendix Table 8 and Figure 4). Shared-cue pairs are 0.19% of all pairs and p99.9 cuts at 0.1%, so p99.9 is a quantile inside the shared-cue population and its value is set by that population's size as much as by its geometry; §4.3 measures the population directly. Word-cluster bootstrap CIs (B=1000) are ±0.03° at p50 and ±0.3° at p99.9, inside the seed spread; a delete-one-block jackknife gives ±0.1° and ±1°, comparable to it.

Width buys dimensions, not accuracy. The registered sweep held the corpus, steps and learning rate (10^{-4}, tuned at d=4096) fixed at d=512–4096 (seed 0, bf16; registered 2026-09-02) and found accuracy rising from 0.722 to 0.908. That trend was the learning rate. Extending the sweep down to d=64 (registered 2026-09-05, three seeds) the small models did not learn at 10^{-4}, and at the toy-arm rate 10^{-3} they reach 0.872 at d=64 and 0.908–0.935 at d=128–256, above the d=4096 model. Re-running the registered widths at 10^{-3} lifts d=512 to 0.902 and d=1024 to 0.815 and diverges at d=2048, and 3×10^{-4} lowers d=4096 to 0.867; the optimal rate falls with width. Taking the better of the rates tried, accuracy is 0.87 at d=64 and 0.90–0.93 from d=128 up (Figure 2, left; d=1024 had neither rate near optimum). The task saturates at about 0.9 with 128 dimensions.

The code does not saturate. Its participation ratio is 8 at d=64, 10 at 128, 16 at 256 and 50 at 4096, about $d^{0.45}$. Noise does not explain the growth: splitting each sense's sentences in two and taking the cross-half covariance, which is unbiased for the signal, gives signal PR 7.4 / 9.8 / 15.5 / 22.9 / 42.1 at d=64 / 128 / 256 / 512 / 4096, with noise 9–32% of the trace (Figure 2, right; the 1024 and 2048 points, 33 and 26, come from the two runs where neither rate was near optimum). So SGD spends about seven dimensions on the within-word code when it has 64 and about forty when it has 4096, for the same accuracy. The cross-word bulk tracks the spectrum-matched null at every width (p50 15.5° against 14.8° at d=64, 5.44° against 5.37° at 4096), and the shared-cue tail is width-invariant (median 68°–73° at d=64–256, 55° at 4096; p99.9 55°–73°) while the null's tail falls as PR grows. The registered predictions for the low-width sweep, spread / squash-and-collapse / drop the rare senses, all failed; the model squashes and succeeds.

Table 2: Shared-cue alignment before and after training (seed 0, post-hoc, nulls.py). Medians in degrees; PR is the participation ratio of the protocol centroids; Jaccard bins are the four cue-overlap bins of Figure 3. This is a separate capture (training rows, fp32 forward, words with a missing sense dropped), so its trained values sit about 2° below the main pipeline's (55.4°, Table 3); the init-to-trained comparison is within one pipeline.

	PR	shared	disjoint	ratio	P(dev>21°)	Jaccard bins
main, init	383	17.2	1.64	10.5	0.306	7.7 / 11.7 / 15.8 / 21.1
main, trained	48	53.5	5.55	9.6	0.997	41.2 / 45.6 / 50.9 / 59.7
mixed, init	396	15.1	1.63	9.3	0.179	7.7 / 11.9 / 14.7 / 18.4
mixed, trained	62	49.2	4.76	10.3	0.998	43.4 / 45.9 / 48.5 / 53.3

4.3 Where the tail lives: inherited cue structure, not co-occurrence

This decomposition was not pre-registered, and it replicates in every run. Pairs that co-occur in sentences (317,619, 0.64%) are not enriched in the tail (mean deviation 7.0° vs 6.5°). Pairs whose senses were trained under a common cue (94,328, 0.19%) have median deviation 55.4° [55.1, 55.6] ($|\cos| \approx 0.82$) and $P(\text{dev} > 21° \mid \text{shared}) = 1.000$ against 0.0067 for cue-disjoint pairs. Deviation is graded in cue-set Jaccard, 43°, 48°, 53°, 62° across the four bins (Figure 3), with seed ranges within 0.6°. Context-disjoint pairs sit at the bulk, median 5.43°, p99.9 25.2°. By count the disjoint population still holds most pairs above 21° (3.3×10^5 against 9.4×10^4); cue sharing owns the extreme tail, not every pair beyond the bulk.

The alignment is present before training. A one-layer model with no positional code sums the prefix embeddings into the residual, so two senses trained under the same cue pool share a cue-embedding component whatever the weights. We measured the same decomposition on the untrained model (seed 0, same corpus; post-hoc, Table 2). Shared-cue pairs already deviate 17.2° at initialization against 1.64° for disjoint pairs, graded in Jaccard 7.7°, 11.7°, 15.8°, 21.1°. After training the two populations sit at 53.5° and 5.55°. The ratio is 10.5× before training and 9.6× after. The pattern, which pairs align and how the alignment grades with cue overlap, is set by the input; training raises its magnitude, shared-cue $|\cos|$ from 0.29 to 0.80, at the same time as it compresses the code (PR 383 to 48) and raises the disjoint bulk in step. SGD does not originate this structure and does not remove it; the mechanism, cue embeddings summed into the residual, is inferred from the architecture and not isolated by an intervention (§6).

Rank-mixed control. In the main corpus every shared-cue pair is same-rank, since the cue pools are indexed by (block, rank). We registered a control that breaks this, one cue pool per block shared by all its ranks, 499 blocks of about 20 mixed-rank senses, so a cue token carries no rank information (verified, 0.61% conflicts, no leakage; registered 2026-09-02, Table 3). Shared-cue pairs deviate 50.1°–50.6°, disjoint pairs 4.76°–4.77°, $P(\text{dev} > 21°) = 1.000$ shared against 0.0028–0.0030 disjoint, and null B p99.9 is 23.2° against observed 49.6°–50.1°. Accuracy is 0.909–0.915, the same as the main corpus. Within the mixed corpus, same-rank shared pairs sit at 54.2° and cross-rank shared pairs at 48.5° (seed 0; 17.2° and 14.8° at initialization), so frequency rank adds about 5°, cue identity carries the rest, and the rank component is present at initialization too; that 5° is the gap between the two corpora. The registered rule had two parts. Part (a), alignment survives, is met. Part (b), that the rank-mean subspace separates from the shared-cue subspace at k=1 ($\cos < 0.7$), failed (cos 1.000). We read the failure as a property of the basis construction, since PCA of pair means puts the frequency axis first in any corpus whose pairs differ in rank composition, but that reading is post hoc and the rule as registered was not passed (Appendix A).

4.4 Pricing interference does not remove it

λ=1 raises the centroid participation ratio 46–50 to 72–77 (top-16 share 45–47% to 33–35%) while lowering the unembedding's from 1847–1865 to 1424–1500. It spreads the targets and

Table 3: Rank-mixed cue pools, one pool per block of about 20 mixed-rank senses, so a cue token carries no rank information. γ=0, bf16, 3 seeds, registered 2026-09-02.

seed	acc	PR	p50	p99.9	null B p99.9	shared med	disjoint med	P(dev>21°) shared / disjoint
0	0.909	62	4.78	50.1	23.2	50.6	4.77	1.000 / 0.0030
1	0.915	62	4.77	49.6	23.2	50.1	4.76	1.000 / 0.0028
2	0.915	62	4.77	49.7	23.2	50.2	4.76	1.000 / 0.0029

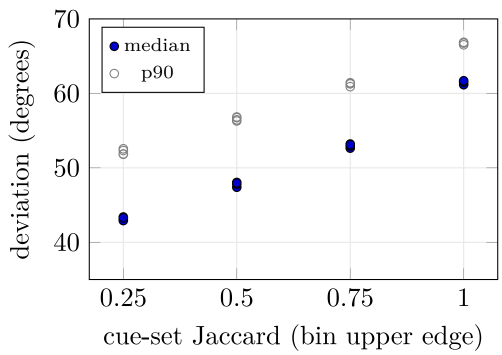

Figure 3: Shared-cue pair deviation against cue-set overlap, four γ=0 seeds overlaid (bin counts 5985/18448/26993/42902). Alignment is monotone in shared-context fraction.

concentrates the reader. It lowers the bulk, p50 5.41°–5.65° to 4.21°–4.39°, and null B falls with it, 5.36°–5.51° to 4.38°–4.52°, ratio to null B 0.96–0.97, so the bulk carries no structure beyond its dimensionality under the penalty as it does without. The tail moves a little and stays. In absolute terms the shared-cue median goes from 55.4° to 53.3°–53.6°, p99.9 from 54.6° to 52.5°–53.0°, the maximum from 73° to 71°–72°, and each Jaccard bin drops about 2°; $P(\text{dev} > 21° \mid \text{shared})$ is still 1.000 under co-activation pricing and 0.999 under uniform pricing, while the disjoint 21° tail falls 0.0067 to 0.0009–0.0010. The ratio of p99.9 to null B rises from 2.0–2.1× to 2.4–2.5× only because the null fell with the bulk, so we do not read it as a strengthening. Weighted interference falls 0.0235 to 0.0150–0.0153, a 35% drop at the medians, and held-out accuracy 0.904–0.915 sits inside the γ=0 range 0.900–0.910; with one seed per pricing this rules out a large cost, not a small one. Under co-activation pricing the tail's survival is expected, shared-cue pairs rarely co-occur and are cheap. Uniform pricing charges every pair type the same expected price, and the shared-cue median still moves only to 53.3°. Given §4.3, this is the inherited cue structure resisting a penalty of this size, not evidence that the task needs it; the registered "load-bearing" criterion is met in letter and we no longer draw its conclusion.

4.5 Readout deletion, and what the fairness check showed

From training rows only, each shared-cue pair contributes its normalized pair mean, and PCA of these gives the shared-cue subspace. Held-out behavior is re-evaluated with the projection onto the top-k directions removed from every residual before the unembedding. The registered predictions held where they should: rare-rank accuracy degrades with k (P1), r5 from 0.62–0.66 to 0.35–0.41 at k=16 while r1 improves 0.983 to 0.991; the centered same-word cosine is unchanged (P3, the identity either way); P2 split, the generic top-PCA also hurt the rare ranks; a random 32-dimensional subspace changes nothing. The fairness check (registered 2026-09-02, Table 4; Appendix Figure 5) then failed its rule in all five checkpoints. The rank basis at k=1 is the shared-cue direction (cos 0.999–1.000, identical damage); the cue basis coincides with the shared subspace from $k \geq 4$ (cos 0.98–0.99), the generic PCA from $k \geq 8$ (0.95–0.99), the rare-centroid basis from $k \geq 16$ (0.94–0.99), with damage within the seed spread in each case. The FAIL branch was honored and the initial reading, an aligned subspace set apart as a channel of disambiguation, is retracted.

Table 4: Deletion-fairness controls, γ=0 seed 0, r5 held-out accuracy after deleting k directions, with the median principal-angle cosine to the shared-cue basis at the same k. rank spans the 5 per-rank means. cue is the PCA of per-cue-token means. randvar is a random subspace of k' columns matched to shared-k=16's removed variance. All five checkpoints in Appendix Table 9.

basis	k	r5	cos to shared
baseline	0	0.617	—
shared	1	0.586	—
rank	1	0.588	1.000
shared	2	0.483	—
rank	2	0.485	0.972
shared	16	0.353	—
cue	16	0.333	0.994
randvar	1656	0.582	—

What survives is directional and weak. Deleting the top-variance directions hurts the rare senses more than a variance-matched random subspace, r5 0.35–0.41 after 16 shared directions against 0.58–0.63 after removing the same variance from 1018–1670 random directions, and the rank-mixed corpus shows the same gap (at 39.5% removed variance, acc 0.877 shared against 0.900 random). But 16 directions inside a 50-dimensional code and 1600 directions in 4096 are different operations, every corpus-informed basis coincides once k is large enough, and deleting k=1 alone is seed-inconsistent on r5 (+0.03, −0.035, −0.02). No basis is identified and no mechanism is named.

5 Related work

Elhage et al. (2022) and Scherlis et al. (2022) give the packing vocabulary and the capacity-allocation law. For independent sparse features their saturation objects are tight frames, and their treatment of correlated features allows collapse or alignment, which is the regime of same-word senses and of shared-cue pairs here. Lecomte et al. (2023) propose polysemanticity from initialization and dynamics; our shared-cue alignment is present at initialization and preserved by training, which is a different mechanism, the input embedding rather than dynamics. Liu et al. (2025) make weight decay the superposition knob and place production models in the strong-superposition regime; our γ window replicates the qualitative effect in a one-layer language model, and λ is a knob they lack. Ivanov et al. (2026) prove that capacity saturation forces tight frames in toy models with independent features. Our features are correlated and our model is not shown to saturate, so their theorem is not what we test; what we observe is that SGD moves away from the Gaussian spectrum toward a code of PR 46–50, about 1.2% of the width, and that the per-word-centered cosine that reads as a simplex is an identity of the estimator. Garg et al. (2026) bound how many features linear readout admits, with near-orthogonal random constructions achieving the upper bound; our setting complements the bound, and SGD does not land on the extremal construction. Bereska et al. (2025) measure effective features via sparse-autoencoder (SAE) entropy; we measure them from labelled senses without a dictionary. Svozil (2025) states the near-orthogonality capacity and readout-interference limit for polysemy conceptually; this is the empirical test. Prior work measures polysemy geometry in trained models directly. Arora et al. (2018) model senses as sparse atoms. Coenen et al. (2019) and Ethayarajh (2019) measure BERT-family geometry and anisotropy, which is why we center per word. Minegishi et al. (2025) locate disentanglement in depth, not in a variable at one layer. Engels et al. (2025) find non-linear features, and our linear readout deletion fails to name a basis (§4.5), which a code that is not linearly separable into named parts would produce. Ait Hou & Hwa (2026) identify the lexical-identity confound that per-word centering removes by construction. On interventions, Miller et al. (2026) regularize an SAE dictionary toward orthogonality for isolated interventions; we regularize the model's residuals during training and find a population the regularizer cannot flatten. Jermyn et al. (2022) engineer monosemanticity in toy models. Duan (2026) audits feature survival under quantization; we do not.

The delta, relative to Ivanov et al. (2026) and Elhage et al. (2022), is the object and the regime. Their predictions concern independent sparse features at saturation; here a next-token model is trained on a language with correlated, Zipf-distributed senses whose inventory exceeds the width, and the question is what SGD does with the near-orthogonal headroom it has. It compresses the code to about 1% of the width, the low-rank bias of gradient training (Papyan et al., 2020; Ansuini et al., 2019; Huh et al., 2023) seen in a setting where the feature inventory is known, and the only structure in the code beyond its dimensionality is one the input embeddings already contain, which training amplifies and a pricing penalty does not remove. We know of no prior next-token experiment with a known sense inventory exceeding the width and known co-activation, which is the delta here; Elhage et al. (2022) and Scherlis et al. (2022) test correlated features in autoencoders.

6 Limitations

One layer, one head, no positional code, one architecture, one machine, one author. The prefix is a bag, which at toy width produces a 17% order-blind conflict rate (§3) and at capacity 0.6%. The pre-registered three-way rule on the per-word-centered cosine was ill-posed, since the centering forces $-1/(k-1)$ for exchangeable senses; the -0.25 values in the tables are identity checks. The initialization baseline of §4.3 is one seed on each corpus and was run after the second review round; the mechanism it points to, cue embeddings summed into the residual, is inferred from the architecture and not isolated by an intervention such as ablating the cue tokens' attention contribution. The shared-cue subspace is a PCA of pair means, which puts the frequency axis first in any corpus; part (b) of the rank-mixed rule failed and our reading of that failure is post hoc. The main corpus has rank-pure cue pools; the rank-mixed corpus (three seeds, one block count) is the mitigation. Geometry is measured on seen data, and r_4, r_5 centroids have split-half reliability 0.56, 0.49, so their noise inflates PR toward the Gaussian value and 46–50 is an upper bound on the code. The λ arms are single-seed. The width sweep tried two learning rates per width (three at d=4096), not a tuned schedule, so the d=1024 accuracy is low for that reason and the split-half PR is one seed per width. The decomposition of §4.3 and the nulls of §4.2 were not pre-registered; replication across all runs is the mitigation. None of this transfers to production models by itself.

7 Conclusion

When senses outnumber dimensions, SGD spends the width on accuracy and compresses the code to 46–50 of 4096 effective dimensions; the bulk of the code's non-orthogonality is that dimensionality. The far tail is the cue structure of the input, aligned before training and amplified by it in proportion to the bulk, with most of it on cue identity and about 5° on frequency rank. Width buys dimensions the task does not use, the code growing as $d^{0.45}$ at constant accuracy, and leaves the tail flat; interference pricing spreads the bulk and lowers the tail by 2°; readout deletion identifies no basis. Every failed rule is reported. The near-orthogonal headroom the capacity argument makes available is left unused, and nothing in the code beyond its dimensionality originates in training.

Reproducibility and AI use. Code, corpus generator, pre-registration text, run logs, saved centroids and the table-generation script are at https://github.com/negoro26/simplex-not-orthogonal; every table cell comes from stats.py on the saved artifacts, and Appendix A dates each registered item (the repository is a snapshot, so the dates are the author's record). A capacity run takes $\approx$ 15 minutes on an 8 GB RTX 5060 laptop GPU. An AI coding assistant (Claude, Anthropic) was used under the author's direction for implementation, analysis scripting, literature search and drafting; the author verified every number, citation and claim.

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A Pre-registration ledger

Fairness decision rule (paraphrased, registered before execution, full text in the ablate.py header). PASS, at matched k and matched removed variance, shared's r_5/r_4 damage exceeds every control by more than the γ=0 seed spread, and median $\cos(\text{shared}, \text{rank} \mid \text{cue})$ at k=16 is < 0.7. FAIL, the rank or cue basis reproduces the shared damage curve within seed noise, stop, do not run the planned penalty-concentration retrains, reframe as "SGD builds a shared context code instead of an orthogonal code". PARTIAL, shared beats rank/cue on r_5 but overlaps them ($\cos \geq 0.7$), the set-apart channel reading is dropped.

B Numerics amendments and gates

All registered before the first run that used them; none introduces a new hyperparameter. (1) lr $10^{-3} \rightarrow 10^{-4}$ after 300-step divergence probes at d=4096. (2) TF32 for training and capture matmuls; statistics-critical Gram matrices strict fp32 on CPU. (3) Chunked cross-entropy, activation checkpointing, fused AdamW for the 8 GB fit; loss/grad relative delta 7.7×10^{-8} / 2.8×10^{-7}. (4) Gather loss-bearing rows before the unembedding, drop block checkpointing; bitwise on the gradient fingerprint; reanalysis of a saved checkpoint reproduces the stored artifact bitwise. (5) bf16 autocast with fp32 master weights and fp32 penalty statistics; gates: one-step loss delta 2×10^{-7}, max grad-norm delta 1.8×10^{-3}, and a full bf16 seed-0 run whose per-rank split-half reliability is within 0.006 of the TF32 run on every rank. (6) Pad skipping in per-token GEMMs (22% of positions are padding; attention is causal with no positional code, so pad outputs are dead); loss and residuals bitwise identical, grad-norm delta $\leq 8 \times 10^{-7}$. Step time went 0.435 → 0.153 s over (3)–(6) on the same GPU.

C Additional tables and figures

D Corpus diagnostics

Capacity corpus, seed 0 (check_gen.py): 96,000 sentences of length 8/11/14 tokens ($\approx$ one third each), 9,600 held out. (cue, word)→continuation is a function on all 34,179 observed pairs; each word has exactly 5 continuations; each cue 8–20. Rank frequencies 0.438/0.229/0.146/0.104/0.083; prior-only ceiling 0.438; chance 10^{-4}. Order-blind conflicts

Table 5: Registered items, date written down, outcome. Full text in DISCOVERY.md and the ablate.py header.

item	registered	outcome
Toy protocol: decision rule ($\pm$0.15), per-word centering, $\geq$ 5 seeds, no LayerNorm, untied embeddings	2026-08-22	held throughout
Arm-5 design: corpus, d=4096, endpoints 1–4, ε-grid, matched-random baseline	2026-08-22	run as registered
Amendments 1–6 (numerics; Appendix B)	08-22 to 09-02	each gated before use
Capture deviation: geometry on full corpus, behavior held-out	2026-08-23	stated in §2
λ arm at capacity (coact)	2026-08-23	-35% interference (medians); shared-cue median 55.4° $\rightarrow$ 53.6°
Uniform-pricing fork: "unpriced" vs "load-bearing" criterion	2026-08-24	alignment survives
Readout ablation P1–P3	2026-08-24	P1, P3 confirmed; P2 split
Fairness controls + PASS/FAIL/PARTIAL rule	2026-09-02	FAIL: rank k=1 $\equiv$ shared k=1 in 5/5; set-apart channel reading retracted
Post-hoc (not registered): shared-cue / co-occurrence split, Jaccard dose-response	2026-08-23	replicated in all runs
Width sweep (same corpus and steps, d=512–4096, seed 0, bf16)	2026-09-02	accuracy arm fired (0.722–0.908), later shown to be the fixed learning rate; PR 13–50; tail width-invariant
Low-width sweep $d \in \{64, 128, 256\}$, 3 seeds, predictions spread / squash / drop, lr caveat	2026-09-05	none occurred: acc 0.87–0.93, PR/d 0.06–0.13; caveat fired, 10^{-3} reruns count
Rival hypotheses for PR growth: split-half signal PR, lr-confound runs	2026-09-05	signal PR 7.4 $\rightarrow$ 42 (5.7$\times$, threshold 3$\times$): growth is real; noise 11–32% of trace; accuracy flat at best-of-lr, "capacity is binding" withdrawn
Rank-mixed cue pools (499 blocks, mixed-rank pools, 3 seeds)	2026-09-02	part (a) met, alignment survives (shared 50.1°–50.6° vs 4.8°); part (b) failed (cos 1.000), re-scored post hoc as a property of the basis construction; rule as registered not passed
Post-hoc (not registered): centering identity, spectrum-matched null B, isotropic-in-PR-dim prediction	2026-09-02	stated in §4.1 and §4.2
Post-hoc (not registered): shared-cue alignment on the untrained model, both corpora	2026-09-05	present at init (17° vs 1.6°), preserved in ratio by training; "SGD builds a context code" withdrawn (§4.3)

(a distractor cue that pairs with the target word for a different continuation): 600 sentences (0.62%), best-guess ceiling $\geq$ 0.993 per rank. Zero duplicate sentences, zero test sentences present in train. 4 of 814 held-out r_5 rows have a sense with no training example (reachable ceiling 0.995); 9 of 9,999 senses have fewer than 2 training examples. Toy corpus: conflicts 3,258/19,200 (17%), ceiling 0.917 overall (1.0/0.88/0.82/0.79 by rank under a most-frequent tiebreak); observed r_4 accuracy 0.89–0.91 exceeds the naive ceiling, so the model resolves some conflicts through the distractor word's presence in the bag.

Example sentence: <bos> c410_1_3 w82 f51 c0_1_2 w0 cont_w0_1 f61 c166_2_1 w733 f64.

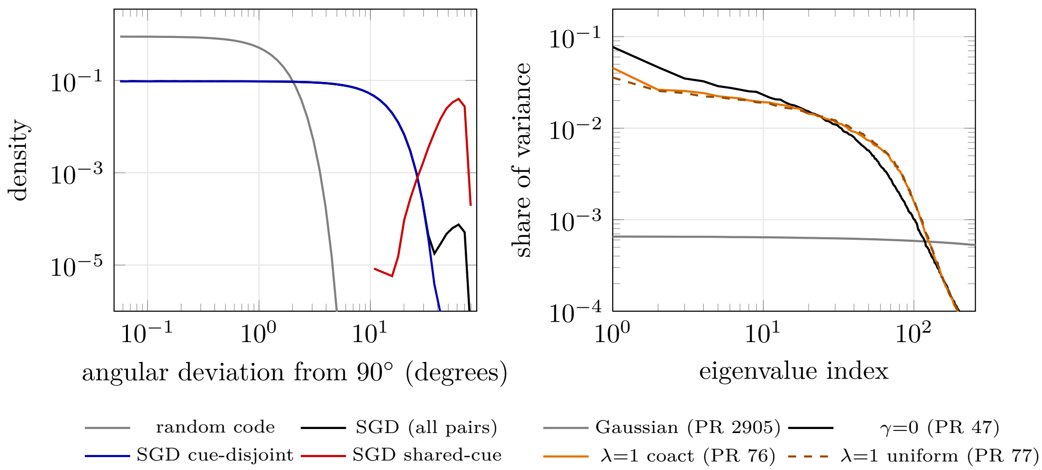

Figure 4: Left, cross-word angular deviation from 90°, seed 0, 5.0×10^7 pairs. The random code sits at 0.6°, the SGD bulk at 5°–6°, and the shared-cue population (0.19% of pairs) at 43°–62°. Right, frame-operator spectrum of the centered sense centroids (top 256 of 4096 eigenvalues). The Gaussian reference is flat at $\approx 1/4096$, SGD concentrates 45%–47% of variance in 16 directions, and the penalty flattens the top of the spectrum.

Table 6: Toy regime, d=64, 5 seeds per row. Each cell: median, with the seed range beneath. Same-word cosines are means over rank pairs; the regular-simplex value is −0.333. Merge: runs in which r_3–r_4 is the least-negative same-word pair (chance 1/6). γ=4 is the death regime; its geometry is void by protocol (reliability floor 0.39).

arm	acc	interference	r_3–r_4 cos	other pairs cos	merge
γ=0	0.911 [0.898, 0.926]	0.318 [0.316, 0.361]	−0.045 [−0.348, +0.094]	−0.380 [−0.405, −0.322]	4/5
γ=1	0.909 [0.902, 0.914]	0.347 [0.322, 0.394]	−0.111 [−0.262, +0.078]	−0.369 [−0.407, −0.343]	5/5
γ=1.5	0.908 [0.901, 0.912]	0.368 [0.354, 0.436]	−0.080 [−0.137, +0.162]	−0.372 [−0.424, −0.368]	5/5
γ=2	0.894 [0.874, 0.903]	0.386 [0.356, 0.460]	+0.022 [−0.158, +0.372]	−0.392 [−0.467, −0.357]	5/5
γ=3	0.882 [0.825, 0.903]	0.416 [0.331, 0.481]	+0.128 [+0.052, +0.408]	−0.410 [−0.476, −0.400]	4/5
γ=4	0.838 [0.677, 0.883]	0.400 [0.395, 0.430]	(death regime)		3/5
λ=1	0.909 [0.898, 0.926]	0.246 [0.232, 0.259]	−0.313 [−0.460, −0.257]	−0.330 [−0.340, −0.301]	0/5
λ=3	0.918 [0.897, 0.920]	0.185 [0.162, 0.269]	−0.298 [−0.384, −0.178]	−0.335 [−0.359, −0.314]	0/5

Table 7: Frame-operator spectrum (F=$W^\top W$) of the centered 10^4 sense rows of the unembedding and of the centered sense centroids. PR =$(\sum \lambda)^2 / \sum \lambda^2$; Gaussian reference (same shape) 2905.

	unembedding rows			sense centroids		
run	PR	λ_1 share	top-16 share	PR	λ_1 share	top-16 share
γ=0, seeds 0–3	1847–1865	1.1%	3.7–3.8%	46–50	7.0–8.4%	45–47%
λ=1 coact (2 runs)	1424–1428	1.7%	4.2%	72–76	3.8–4.6%	34–35%
λ=1 uniform	1500	1.6%	4.2%	77	3.6%	33%

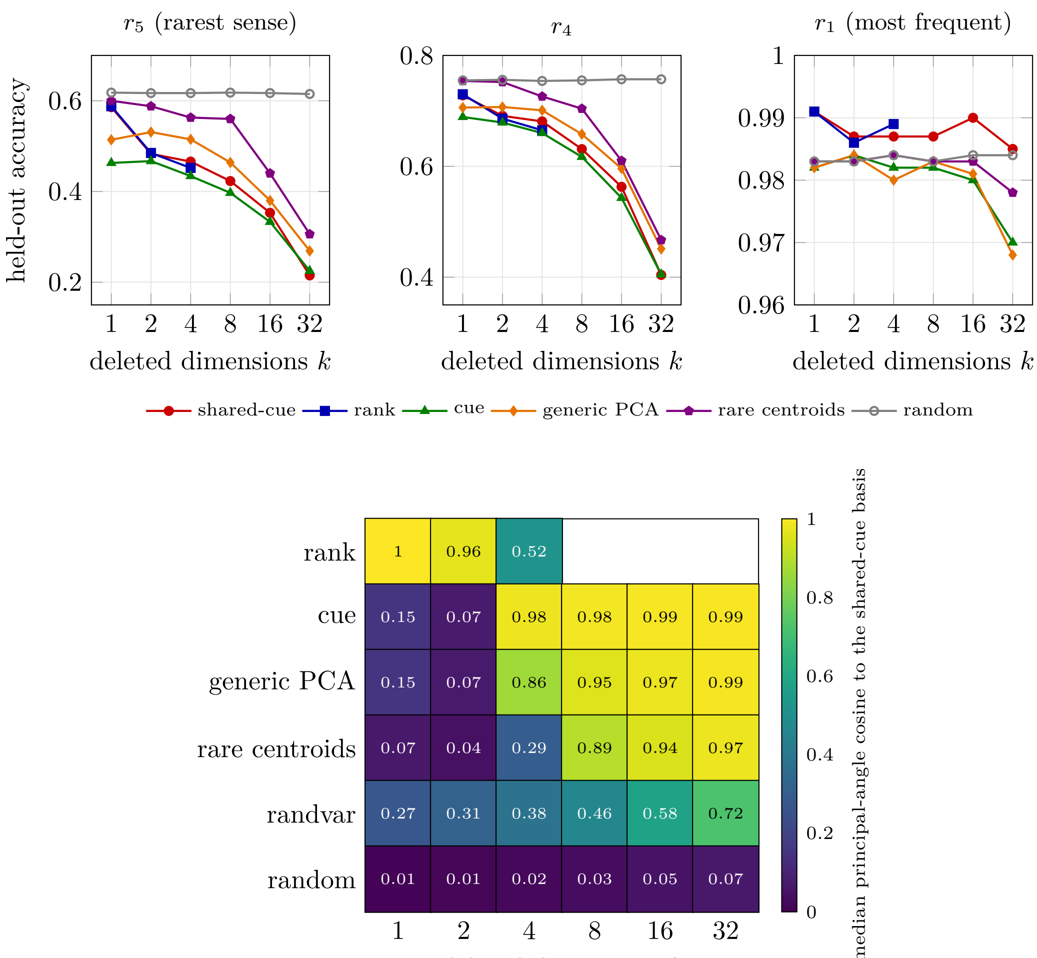

Figure 5: Readout deletion, γ=0 seed 0. Top: held-out accuracy after removing the top-k directions of each basis (rank spans only 5 directions). Damage lands on rare senses and every rank-informative basis produces it; a random basis is inert. Bottom: overlap between each control basis and the shared-cue basis at matched k, mean over the five checkpoints. rank at k=1 is the shared-cue direction (0.999); cue and generic coincide with it from k=4 on. randvar is the variance-matched random subspace ($k' \gg k$).

Table 8: Cross-word deviation, nulls, and cluster 95% CIs, per run. Null A, isotropic unit vectors in $\mathbb{R}^d$, is the paper's original baseline. Null B, a spectrum-matched Gaussian ($N(0, \Sigma)$ with the empirical centroid covariance, normalized), matches the bulk. d_{eff}, the median deviation of isotropic vectors in PR dimensions. Each CI cell: estimate, word bootstrap CI (B=1000, 2000 clusters), delete-one-block jackknife CI (500 blocks of 4 words sharing a cue pool; each deletion removes about 2×10^5 pairs, a delete-d jackknife with $d \gg \sqrt{n}$, which is consistent for quantiles). A block bootstrap is not used: it gives within-block pairs weight m_b^2 with expectation 2 against 1 for cross-block pairs and pulls the tail quantile toward the shared-cue population.

run	PR	p50 word / block CI	p99.9 word / block CI	null A p50 / p99.9	null B p50 / p99.9 / max	d_{eff} p50
γ=0 s0 fp32	47	5.59 5.57–5.62 5.48–5.71	54.2 53.9–54.6 53.2–55.3	0.60 / 2.94	5.49 / 26.4 / 44.7	5.63
γ=0 s0 bf16	50	5.44 5.42–5.47 5.33–5.55	54.6 54.4–55.0 53.7–55.5	0.60 / 2.94	5.37 / 25.9 / 43.2	5.49
γ=0 s1 fp32	50	5.43 5.40–5.45 5.33–5.52	54.8 54.5–55.1 53.9–55.7	0.60 / 2.95	5.39 / 25.8 / 42.4	5.49
γ=0 s2 fp32	46	5.65 5.62–5.68 5.52–5.78	54.5 54.2–54.8 53.6–55.3	0.60 / 2.95	5.51 / 26.8 / 42.9	5.68
γ=0 s3 fp32	50	5.41 5.39–5.44 5.31–5.51	54.7 54.4–55.0 53.8–55.7	0.60 / 2.94	5.36 / 25.8 / 44.7	5.47
λ=1 coact s0 fp32	76	4.24 4.23–4.25 4.20–4.28	52.8 52.5–53.1 51.8–53.7	0.60 / 2.94	4.41 / 21.1 / 35.3	4.43
λ=1 coact s0 bf16	72	4.39 4.38–4.40 4.34–4.44	53.0 52.7–53.3 52.1–53.8	0.60 / 2.94	4.52 / 21.7 / 35.5	4.56
λ=1 uniform s0 bf16	77	4.21 4.20–4.22 4.17–4.24	52.5 52.2–52.8 51.5–53.5	0.60 / 2.94	4.38 / 21.0 / 34.0	4.40

Table 9: Fairness controls on all five checkpoints. Each cell: r_5 held-out accuracy after deleting k directions of the shared-cue basis / of the control basis, with the median principal-angle cosine between the two bases beneath. Last column: the variance-matched random subspace at shared-k=16's energy, as k' columns and the resulting r_5. Baseline r_5: 0.617, 0.643, 0.625, 0.658, 0.625.

run	shared / rank, k=1	shared / rank, k=2	shared / cue, k=16	randvar k', r_5
γ=0 s0	0.586 / 0.588 cos 1.000	0.483 / 0.485 cos 0.972	0.353 / 0.333 cos 0.994	1656, 0.582
γ=0 s1	0.634 / 0.643 cos 0.999	0.571 / 0.548 cos 0.967	0.389 / 0.465 cos 0.994	1623, 0.606
γ=0 s2	0.634 / 0.643 cos 0.999	0.545 / 0.534 cos 0.990	0.373 / 0.360 cos 0.994	1670, 0.589
λ=1 coact	0.559 / 0.559 cos 0.999	0.498 / 0.486 cos 0.958	0.367 / 0.345 cos 0.982	1049, 0.630
λ=1 uniform	0.607 / 0.608 cos 0.999	0.531 / 0.505 cos 0.924	0.405 / 0.407 cos 0.982	1018, 0.601